

3D Product View — Implementation Plan

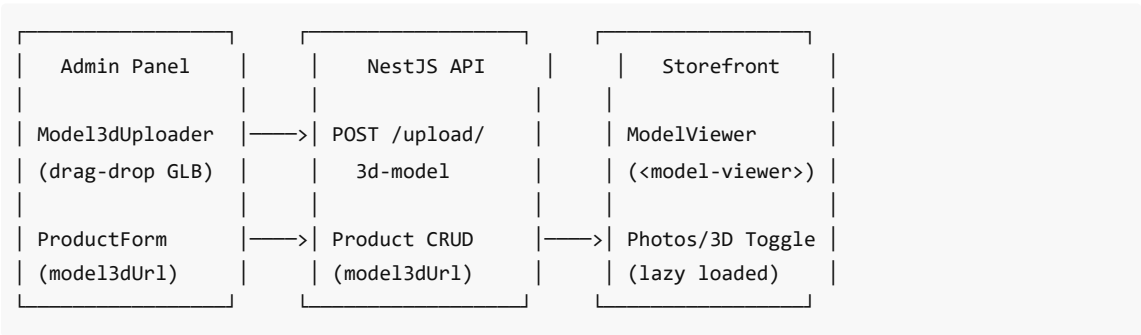
Overview

This document describes the implementation of interactive 3D product viewing on the Naro Fashion e-commerce storefront using Google's open-source `<model-viewer>` web component. The feature allows customers to rotate, zoom, and on mobile use AR ("View in your space") to interact with product 3D models.

Why `<model-viewer>`

Feature	Detail
Cost	Free, open-source (Google)
Formats	GLB/GLTF (industry standard)
Interaction	Rotate, zoom, pan (mouse + touch)
AR Support	Built-in — Android Scene Viewer, iOS Quick Look
Bundle Size	~100KB gzipped, lazy-loadable
Browser Support	All modern browsers + mobile Safari/Chrome

Architecture



Phase 1: Database & API

Database Schema Changes

Two new nullable fields added to the `Product` model:

```
model Product {
  // ... existing fields ...

  model3dUrl      String? // URL to GLB/GLTF file in /uploads/models/
  model3dPosterUrl String? // Optional poster image while 3D loads

  // ... rest of model ...
}
```

API Endpoints

Method	Endpoint	Description	Limit
POST	/upload/3d-model	Upload GLB/GLTF file	25MB
POST	/products	Create product (includes model3dUrl)	—
PATCH	/products/:id	Update product (includes model3dUrl)	—

Upload validation:

- Accepted MIME types: `model/gltf-binary` , `model/gltf+json` , `application/octet-stream`
- Also checks file extension: `.glb` , `.gltf`
- Max file size: 25MB
- Storage: `uploads/models/` directory

Product DTOs

Both `CreateProductDto` and `UpdateProductDto` include:

```
@IsOptional()
@IsString()
model3dUrl?: string;

@IsOptional()
@IsString()
model3dPosterUrl?: string;
```

Phase 2: Admin Dashboard

Model3dUploader Component

File: `apps/admin/components/products/Model3dUploader.tsx`

A drag-and-drop upload component for 3D model files with:

- Drag-and-drop zone accepting `.glb/.gltf` (max 25MB)
- Upload progress spinner
- Live `<model-viewer>` preview after upload (camera controls + auto-rotate)
- Remove button to clear the model
- Toast notifications for errors/success

Props:

```
interface Props {
  modelUrl: string | null;
  posterUrl: string | null;
  onModelChange: (url: string | null) => void;
  onPosterChange: (url: string | null) => void;
}
```

ProductForm Integration

The `Model3dUploader` is rendered in a new "3D Model (Optional)" section between the Images and Variants sections of the existing `ProductForm`.

Phase 3: Storefront

ModelViewer Component

File: `apps/storefront/components/product/ModelViewer.tsx`

A client component that dynamically imports `@google/model-viewer` and renders an interactive 3D viewer with:

- Camera controls (mouse/touch rotate, zoom, pan)
- Auto-rotate when not interacting
- AR support on mobile devices
- Lazy loading (model downloads only when viewer is visible)
- Shadow and lighting for realistic presentation

Product Detail Page Toggle

When a product has a 3D model (`model3dUrl` is set), the product detail page shows two toggle buttons above the image gallery:

Button	Icon	Description
Photos	Image icon	Shows the existing photo gallery (default)
3D View	Box icon	Shows the interactive 3D model viewer

The 3D viewer is lazy-loaded via `next/dynamic` with `ssr: false` — it only downloads when the user clicks "3D View", ensuring zero impact on page load speed.

Poster image: While the 3D model loads, a poster image is shown. If no dedicated poster is set, the product's primary photo is used as fallback.

AR on mobile: The `<model-viewer>` component automatically shows an AR button on supported devices:

- **Android:** Google Scene Viewer (Chrome)
- **iOS:** Apple Quick Look (Safari, iOS 12+)

Content Creation Guide

How to Create 3D Models from Product Photos

Recommended Free Tools

Tool	Platform	Method	Best For
Luma AI	Web (lumalabs.ai/genie)	Upload 8-20 photos → AI generates 3D	Gowns on mannequins
Polycam	iOS / Android app	Photo or LiDAR scan → export GLB	Quick captures
Meshroom	Windows (NVIDIA GPU)	Photogrammetry from 20-50 photos	High-detail items

Blender	All platforms	Manual 3D modeling → export GLB	Full creative control
Sketchfab	Web marketplace	Download free CC-licensed models	Generic fashion items

Photography Tips for Best 3D Results

- 1. **Lighting:** Use consistent, even lighting — no harsh shadows
- 2. **Background:** Place product on solid white or light gray background
- 3. **Angles:** Take photos at 3 heights — eye level, 45° above, slightly below
- 4. **Coverage:** Walk around the product, taking a photo every 10-15° (20-30 total)
- 5. **Overlap:** Each photo should overlap ~60-70% with the previous one
- 6. **Details:** Include close-ups of fabric texture, embellishments, labels
- 7. **Distance:** Keep consistent distance from the product

File Optimization

GLB files should ideally be under 5MB for fast loading on mobile networks in Tanzania. Use this free tool to compress:

```
npx @gltf-transform/cli optimize model.glb optimized.glb --compress draco
```

Recommended Workflow

- 1. Place the garment on a dress form or mannequin
- 2. Take 20-30 photos walking around it in a circle
- 3. Upload to **Luma AI** or **Polycam**
- 4. Export as **GLB** format (single file with embedded textures)
- 5. Optimize with `gltf-transform` if over 5MB
- 6. Upload via admin dashboard → Product Edit → 3D Model section

Technical Details

Dependencies

Package	Version	Apps
@google/model-viewer	^4.0.0	admin, storefront

Type Declarations

Both admin and storefront need a TypeScript declaration file for the `<model-viewer>` web component:

`types/model-viewer.d.ts` :

```
declare namespace JSX {
  interface IntrinsicElements {
    'model-viewer': React.DetailedHTMLProps<
      React.HTMLAttributes<HTMLElement>, HTMLElement
    > & {
      src?: string;
      alt?: string;
      poster?: string;
      'camera-controls'?: boolean | string;
    };
  }
}
```

```

    'auto-rotate'?: boolean | string;
    ar?: boolean | string;
    'ar-modes'?: string;
    loading?: string;
    'shadow-intensity'?: string;
    style?: React.CSSProperties;
  };
}
}

```

Files Modified/Created

File	Action
packages/database/prisma/schema.prisma	Modified — added 2 fields to Product
apps/api/src/upload/upload.controller.ts	Modified — added 3D model upload endpoint
apps/api/src/upload/upload.service.ts	Modified — added skipMimeCheck parameter
apps/api/src/products/dto/create-product.dto.ts	Modified — added 3D fields
apps/api/src/products/dto/update-product.dto.ts	Modified — added 3D fields
apps/admin/types/model-viewer.d.ts	Created
apps/admin/components/products/Model3dUploader.tsx	Created
apps/admin/components/products/ProductForm.tsx	Modified — added 3D section
apps/admin/lib/api.ts	Modified — added upload3dModel method
apps/storefront/types/model-viewer.d.ts	Created
apps/storefront/components/product/ModelViewer.tsx	Created
apps/storefront/app/products/[slug]/page.tsx	Modified — added Photos/3D toggle

Testing Checklist

- ☐ Upload a sample GLB file via admin product edit form
- ☐ Verify 3D preview works in admin after upload
- ☐ Open product on storefront — verify "Photos" / "3D View" toggle appears
- ☐ Click "3D View" — verify model loads with rotation/zoom
- ☐ Test auto-rotate stops when user interacts, resumes when idle
- ☐ Test on mobile — verify AR button appears (Android/iOS)
- ☐ Test product WITHOUT 3D model — verify no toggle, normal gallery only
- ☐ Edit product → remove 3D model → save → verify toggle disappears
- ☐ Test with large GLB file (15-20MB) — verify upload works within 25MB limit
- ☐ Test poster image fallback (when no dedicated poster, uses primary product image)